

FFA Working Papers

Impact of size and volume on cryptocurrency momentum and reversal

Milan Fičura

FFA Working Paper 3/2023



FACULTY OF FINANCE AND ACCOUNTING

About: FFA Working Papers is an online publication series for research works by the faculty and students of the Faculty of Finance and Accounting, Prague University of Economics and Business, Czech Republic. Its aim is to provide a platform for fast dissemination, discussion, and feedback on preliminary research results before submission to regular refereed journals. The papers are peer-reviewed but are not edited or formatted by the editors.

Disclaimer: The views expressed in documents served by this site do not reflect the views of the Faculty of Finance and Accounting or any other Prague University of Economics and Business Faculties and Departments. They are the sole property of the respective authors.

Copyright Notice: Although all papers published by the FFA WP series are available without charge, they are licensed for personal, academic, or educational use. All rights are reserved by the authors.

Citations: All references to documents served by this site must be appropriately cited.

Bibliographic information:

Fičura M. (2023). *Impact of size and volume on cryptocurrency momentum and reversal*. FFA Working Paper 3/2023, FFA, Prague University of Economics and Business, Prague.

This paper can be downloaded at: wp.ffu.vse.cz

Contact e-mail: ffawp@vse.cz

Impact of size and volume on cryptocurrency momentum and reversal

Authors

Milan Fičura¹

Abstract

We analyse how cryptocurrency size and trading volume impact the momentum and reversal dynamics of their returns. We show that the previously reported weekly return reversal occurs for small and illiquid coins only (t-stat = -7.31), while the large and liquid coins exhibit weekly momentum effect instead (t-stat = 2.33). Long-term returns exhibit reversal effects, which are, however, insignificant for the large and liquid coins. We further analyse the impact of high momentum on future cryptocurrency returns, measured as the distance of previous-week closing price from the k-week high. High momentum has not been analysed on cryptocurrency markets before, and we show it to be a superior predictor of future returns when compared to regular momentum. The distance from the 1-week high predicts negatively future returns of small and illiquid coins (t-stat = -9.03) and positively future returns of large and liquid coins (t-stat = 4.93). The results are highly robust to different settings of the size and liquidity thresholds. We further show that the short-term reversal of small and illiquid coins is driven mostly by their low trading volumes, while the short-term momentum of large and liquid coins is driven mostly by high market capitalizations and to a lower degree by high trading volumes.

AMS/JEL classification: G11, G12, G17

Keywords: Cryptocurrency, momentum, reversal, high-momentum, size, liquidity, asset pricing

1. Introduction

With the rapid growth of cryptocurrency markets, large number of research emerged trying to identify factors governing the cross-section of cryptocurrency returns. Seminal studies on the topic proposed a 3-factor models of cryptocurrency returns with a cryptocurrency market factor, size factor, and either a long-term momentum (Liu et al. 2020), short-term momentum (Liu et al. 2022) or short-term reversal factor (Shen et al., 2020). Later studies proposed further predictors of cryptocurrency returns, such as liquidity (Zhang and Li, 2023), idiosyncratic volatility (Zhang and Li, 2020), higher-order moments (Jia et al., 2021), seasonality (Long et al., 2020), max effect (Li, et al., 2021), downside risk (Zhang et al., 2021), realized signed jumps (Zhang and Zhao, 2021), volatility of liquidity (Lervik, 2022), or the address-to-network ratio, used as a proxy for the cryptocurrency value (Liebi, 2022).

While the ongoing research increased our understanding of the cryptocurrency return dynamics, the results are likely to be influenced by short history of the cryptocurrency market existence, extreme volatility of cryptocurrency returns, low liquidity of most of the coins, and high market concentration,

¹ Milan Fičura, Faculty of Finance and Accounting, Prague University of Economics and Business, Czech Republic, (milan.ficura@vse.cz, corresponding author).

Note: This paper has been prepared under financial support of a grant GAČR 22-19617S “Modelling the structure and dynamics of energy, commodity and alternative asset prices”, which the author gratefully acknowledges.

where 2% of the largest cryptocurrencies correspond to more than 90% of the overall market capitalization (Zaremba et al., 2021). Methods to analyse the cryptocurrency market have also not yet been entirely standardized, and most of the studies thus differ with respect to the analysed time-period, dataset definition (regarding the exclusion of small and illiquid coins) and the method used to compute the portfolio returns (use of Winsorization, use of equally vs. value weighted returns, etc.).

As a possible consequence of that, the literature gives somewhat conflicting results on the momentum and reversal effects on the cryptocurrency market. While the studies of Liu et al. (2022), Jia et al. (2022) and Dobrynskaya (2023) report a statistically significant short-term momentum, manifesting in a positive impact of 1 to 4 week lagged returns on the future 1-week return, studies of Shen et al. (2020) and Kozlowski et al. (2020) report a significant short-term (weekly) reversal. The results on longer-term momentum and reversal are conflicting as well. Grobys and Sapkota (2019) report an insignificant long-term (12-month and 6-month) momentum but a significant monthly reversal, Liu et al. (2022) observe insignificant impact of long-term (above 4 week) returns on future 1-week returns, Dobrynskaya (2023) finds a statistically significant long-term reversal (above 4-weeks), while Liu et al. (2020) find a statistically significant 52-week momentum.

Possible resolution of the momentum/reversal puzzle on the cryptocurrency market offers the study of Zaremba et al. (2021), showing that small cryptocurrencies (98% of all coins, comprising only 10% of the total market capitalization) exhibit significant 1-day reversal, while “large” cryptocurrencies (2% of all coins, comprising 90% of the total market capitalization) exhibit 1-day momentum instead. An indication of a possible relationship between size and momentum and reversal effects is given also by the design of some of the former studies. Studies performed on older datasets, when cryptocurrencies were small (Grobys and Sapkota, 2019, Kozlowski et al. 2020), and the ones that use equal-weighted returns and no minimum market capitalization filter (Shen et al., 2020) tend to report a reversal effect. The ones done on more recent datasets, with minimum market capitalization filters, and value-weighted returns, report instead a momentum effect (Liu et al., 2022, Dobrynskaya, 2023).

In our study, we explore the possible relationship between cryptocurrency size, trading volumes, and the dynamics of their returns in more detail. Unlike Zaremba et al. (2021), we focus on the weekly frequency, as in the studies of Liu et al. (2022) and Shen et al. (2020). We then dynamically split, at the end of each week, all existing coins into two groups, denoted as Large and Liquid and Small and Illiquid, based on a 50 mil. USD market capitalization and 5 mil. of last-week USD trading volume thresholds and analyse each of them separately. As there is insufficient number of large and liquid coins before 2017/06, we restrict the analysis to the period of 2017/06-2022/12.

The results show that the 1-week return reversal, reported in Shen et al. (2020) and Kozlowski et al. (2020), occurs only for small and illiquid coins ($t\text{-stat} = -7.31$), while the large and liquid coins exhibit a momentum effect instead ($t\text{-stat} = 2.33$), consistent with the results of Liu et al. (2022), and Dobrynskaya (2023). Long-term returns exhibit reversal effects, which are, however, insignificant for the large and liquid coins ($t\text{-stat} = -1.91$ for 26-week returns). We further analyse the impact of high-momentum on future cryptocurrency returns (George and Hwang, 2004), which has not been applied to the cryptocurrency market before (as to our knowledge), and show that it is a superior predictor of future returns, with an even larger discrepancy between the two groups. The 1-week high-momentum predicts negatively future returns of small and illiquid coins ($t\text{-stat} = -9.03$) and positively the future returns of large and liquid coins ($t\text{-stat} = 4.93$). We further extend our analysis beyond our baseline size and liquidity thresholds and show that they are highly robust to their alternative settings. We also show that the short-term reversal of the small and illiquid coins tends to be driven mostly by their low trading volumes, while the short-term momentum of the large and liquid coins is driven mostly by their large market capitalizations.

The rest of this paper is organized as follows: Section 2 describes the dataset construction, exploratory analysis and variable construction, Section 3 discusses the empirical results on quantile portfolio sorts, time-series regressions and robustness checks, and the Conclusion concludes the paper.

2. Data preparation

2.1. Dataset and exploratory analysis

Consistently with previous studies, we use price data observed on Coinmarketcap.com (Liu et al., 2022, Shen et al., 2020), which tracks the prices from more than 200 cryptocurrency exchanges and reports the aggregated volume-weighted market prices, aggregated trading volumes and market capitalizations. As per 2023/01, the website reports a total of 20 685 cryptocurrencies with the price time-series available for 18 352 of them (including coins that were delisted in the past and are not tracked anymore). The number of coins whose prices were tracked at the end of each year and the coins whose market capitalization exceeds different absolute thresholds is shown in **Table 1** below.

Table 1 – Number of coins above different market capitalization thresholds at the end of each year

date	total	MCap>100k	MCap>1M	MCap>10M	MCap>50M	MCap>250M	MCap>1B
2013-12-31	66	65	32	12	6	2	1
2014-12-31	525	128	34	11	4	2	1
2015-12-31	559	131	37	6	4	1	1
2016-12-31	637	231	82	23	7	2	1
2017-12-31	1337	942	697	429	224	86	34
2018-12-31	2055	1348	784	228	71	24	11
2019-12-31	2377	1393	826	270	93	33	9
2020-12-31	4001	1895	1276	551	226	83	34
2021-12-31	8624	2431	1818	1083	601	263	108
2022-12-31	8716	2047	1333	684	315	106	40

Note: The table shows number of coins tracked by Coinmarketcap.com at the end of each year from 2013-2022 together with the number of coins that exceed different market capitalization thresholds.

We can see from **Table 1** that the market capitalization of most of the coins is extremely small, with only 5% of all coins (on average over the years) exceeding the 50 mil. USD threshold that is commonly used for the definition of even a micro-cap in the case of stocks. Over 95% of all cryptocurrencies would thus be considered nano-cap if they were stocks.

In the next step we analyse daily volumes of our coin dataset to assess whether a certain level of market capitalization assures sufficient liquidity of the coin. As of our knowledge no previous study analysing the cross-section of cryptocurrency returns applies a filter on trading volumes for the definition of their dataset. Most of the studies apply either no filter on the cryptocurrency size (Shen et al., 2020), use a minimum market capitalization filter, typically set at 1 million USD (Liu et al., 2022, Zhang and Li, 2023, Dobrynskaya, 2023), or a apply a relative filter by selecting the largest N coins at the end of each week (Li et al., 2021, Kozlowski et al., 2020).

Table 2 – Daily USD volume quantiles after the application of different market capitalization filters

Size filter	CoinDays	Volume_5Q	Volume_25Q	Volume_50Q	Volume_75Q	Volume_95Q
MCap >= 1 bil.	85594	\$4,102,618	\$62,848,484	\$226,565,110	\$855,755,535	\$9,580,408,684
MCap >= 250 mil.	115766	\$366,781	\$6,287,943	\$23,466,354	\$70,007,768	\$294,163,045
MCap >= 50 mil.	328874	\$56,875	\$1,035,631	\$4,097,434	\$13,445,567	\$64,169,562
MCap >= 10 mil.	610939	\$2,991	\$117,955	\$592,066	\$2,229,357	\$12,106,262
MCap >= 1 mil.	1242442	\$0	\$4,199	\$45,522	\$241,624	\$1,632,480
MCap >= 100k	1208566	\$0	\$35	\$874	\$13,553	\$185,250
MCap < 100k	5655195	\$0	\$2	\$722	\$37,620	\$924,875

Note: The table shows the number of coin-days and the 5%, 25%, 50%, 75% and 95% quantiles of daily USD volume computed over all coin-days in our dataset, filtered based on different settings of the minimum market capitalization rule (ranging from 100 thousand to 1 billion USD).

We can see from **Table 2** that with the commonly used 1 million USD market capitalization filter, the dataset still contains 5% of coin-days with 0 USD trading volumes and 50% of coin-days with trading volumes under 45 522 USD. Considering that the reported trading volumes are aggregated over large number of cryptocurrency exchanges, it is unlikely that the liquidity of these coins is sufficient for an average market participant to be able to execute trades for the observed end-of-week market prices. Additionally, even with a 50 mil. USD market capitalization filter, the reported USD trading volumes are still under 56 875 USD in 5% of all coin-days.

As the key hypothesis of our paper is that large coins, traded by institutional investors, behave differently than small coins traded by retail (a possibly affected by low liquidity effects), we decide to apply a filter on both, market capitalization and trading volumes to split our dataset. Consistently with most of the previous studies, we remove all coins with end-of-week market capitalization below 1 million USD from our baseline dataset, to make the results even for the “small” coins more comparable to earlier studies. The remaining coins are then divided into two groups: (1) Large and Liquid ones, defined as previous-week market capitalization above 50 mil. USD and previous-week trading volume above 5 mil. USD, and (2) Small or Illiquid ones, defined as previous-week market capitalization between 1 mil. and 50 mil. USD or USD trading volume below 5 mil. USD. To show that our conclusions are robust to different variants of the settings above, we compute them in section 3.3 for 5 alternative variants of the market capitalization split only (ranging from 100 thousand USD to 250 million USD) and 5 alternative variants of the USD volume split only (ranging from 0 to 25 million USD).

2.2. Dataset processing and variable construction

Our initial dataset contains all 18 352 coins available on Coinmarketcap.com on 2023/01, including the delisted and currently untracked coins to avoid possible survivorship bias. Out of these coins, we further remove all coins whose market capitalization did not increase above 100 thousand USD at any point between 2014/04-2023/01 as they will not be used in the analysis. The removal of these small coins decreases the size of the dataset to 4 614 coins in total. Take note that most of the analyses are performed on coins with end-of-week market capitalization above 1 million USD, but we keep the coins with market capitalization above 100 thousand USD in the initial selection, as they will be used in some of the robustness checks.

In the next step, we remove all coins classified as stable coins, mirror coins or wrapped coins. By trying to uphold a peg with respect to the price of another asset (typically USD, EUR, CNY or Gold) stable coins exhibit different dynamics than standard free-floating coins and their inclusion in the dataset might thus bias the results. A similar argument holds for the mirrored coins that typically strive to mirror the price of certain stock which can make their dynamics different from the of other cryptocurrencies, and the wrapped coins, that track the price of another cryptocurrency on a different

blockchain and can thus be viewed as redundant, merely duplicating the observations in our sample. As a result of these rules, 96 coins were removed (see Appendix 2).

In the next step we compute our return predictors.

Momentum $mom_{t,h}$ at the end of week t computed over previous h weeks is defined as:

$$mom_{t,h} = \sum_{t-h+1}^t r_t \quad (1)$$

Where r_t is the logarithmic return in week t .

High momentum $hmom_{t,h}$ at the end of week t computed over previous h weeks is defined as:

$$hmom_{t,h} = \ln(C_t) - \ln(H_{t,h}) \quad (2)$$

Where C_t is the market close price of week t and $H_{t,h}$ is the highest intraday price attained over the past h weeks.

3. Empirical results

3.1. Quintile portfolio sorts

At the end of each week, we divide our coins into the group of (i) Large and Liquid ones, with market capitalization ≥ 50 mil. USD and previous-week trading volume ≥ 5 mil. USD, and (ii) Small or Illiquid ones, with market capitalization between 1 mil. USD and 50 mil. USD or market capitalization > 50 mil. USD but last-week USD trading volume < 5 mil. USD. From both groups we further remove all coins with end-of-week closing price below 0.001 USD in order to avoid possible rounding issues.

The coins in each of the two groups are then sorted based on each of the analysed characteristics (momentum or high-momentum) computed over the past 1-week, 2-week, 4-week, 12-week and 26-week time horizon. The sorted coins are then divided into 5 quintile portfolios, based on each of the characteristics, and the following week return of these portfolios is computed (including a long-short portfolio of quintile 5 minus quintile 1). For the calculation of portfolio returns we use the simple returns (not log-returns), Winsorized with a 1000% maximum return in order to avoid possible outliers and data errors. We compute the portfolio returns as equal-weighted as in Shen et al. (2020) instead of value-weighted as in Liu et al. (2022), as it is more suitable for our analysis of the relationship between the cryptocurrency size and the dynamics of their returns.

Apart from numerical analysis of the long-only and long-short quantile portfolio returns, we provide the charts of the long-short portfolio (Q5-Q1) cumulative returns in Appendix 1. The charts show that the reversal effects observed for small and illiquid coins remains stable over the entire period between 2017/06 the 2022/12. The same holds for the high-momentum observed for large and liquid coins, which is also stable over the entire time-period. In contract to that, the standard momentum seems to have stopped in the early months of 2021 and remains unprofitable since then.

Table 3 – Momentum/Reversal – Average quintile portfolio returns and t-statistics

Momentum/Reversal - Large and Liquid cryptocurrencies							
variable	metric	Q1	Q2	Q3	Q4	Q5	Q5-Q1
mom_1W	mean	-0.20%	1.09%	1.44%	1.33%	1.15%	1.36%
	t-stat	-0.203	1.035	1.290	1.138	0.856	2.332
mom_2W	mean	-0.06%	0.78%	1.35%	1.38%	1.38%	1.44%
	t-stat	-0.064	0.762	1.174	1.197	1.027	2.686
mom_4W	mean	0.34%	1.15%	0.96%	0.99%	1.17%	0.84%
	t-stat	0.340	1.033	0.891	0.857	0.936	1.570
mom_12W	mean	1.32%	1.15%	0.95%	0.96%	0.44%	-0.88%
	t-stat	1.255	0.989	0.870	0.875	0.387	-1.862
mom_26W	mean	1.33%	1.24%	0.89%	0.53%	0.62%	-0.71%
	t-stat	1.248	1.113	0.810	0.512	0.616	-1.914
Momentum/Reversal - Small or Illiquid cryptocurrencies							
variable	metric	Q1	Q2	Q3	Q4	Q5	Q5-Q1
mom_1W	mean	3.44%	1.17%	1.47%	1.45%	-0.47%	-3.91%
	t-stat	3.206	1.079	1.263	1.208	-0.407	-7.309
mom_2W	mean	3.32%	1.15%	1.11%	1.52%	-0.03%	-3.35%
	t-stat	3.351	1.065	0.988	1.198	-0.026	-6.352
mom_4W	mean	3.24%	1.24%	1.28%	1.18%	0.09%	-3.15%
	t-stat	3.255	1.186	1.020	1.021	0.079	-6.769
mom_12W	mean	3.14%	1.73%	1.36%	1.02%	-0.30%	-3.44%
	t-stat	2.978	1.642	1.238	0.867	-0.291	-7.969
mom_26W	mean	2.83%	1.88%	1.54%	0.71%	0.08%	-2.75%
	t-stat	2.571	1.640	1.389	0.788	0.072	-5.606

Note: The table shows average weekly returns (mean) and Newey-West t-statistics (t-stat) of 5 quintile portfolios (Q1 to Q5) and the Q5-Q1 long-short portfolio, with the sorts performed based on the past 1-week, 2-week, 4-week, 12-week and 26-week returns, over the period of 2017/06-2022/12. Instances with t-statistic>2 are shown in bold. The upper table shows the results for Large and Liquid cryptocurrencies with previous-week market capitalization ≥ 50 mil. USD and previous-week trading volume ≥ 5 mil. USD. The lower table shows the results for Small or Illiquid cryptocurrencies with market capitalization < 50 mil. USD (but still > 1 mil. USD) or volume < 5 mil. USD.

We can see from **Table 3** that the Large and Liquid cryptocurrencies exhibit short-term momentum where the 1-week and 2-week returns positively predict the future 1-week return (with t-statistics of 2.33 and 2.87 respectively). Beyond 4-weeks there is a reversal effect that happens to be strongest for the 26-week past returns (t-statistics of -1.94). The results are mostly consistent with the ones reported by Liu et al. (2022) and Dobrynskaya (2023) who work with all coins exceeding the 1-million market capitalization but use the value-weighted return calculation. The results are also consistent with the initial underreaction and delayed overreaction explanation of the momentum anomaly. We can further see that the short-term momentum effect is driven mostly by the short leg of the long-short portfolio and the low returns in the Q1 quintile. Utilization of these results for practical investing is thus dependent on our ability to short the Large and Liquid coins.

For the Small and Illiquid cryptocurrencies, we observe a highly statistically significant reversal effect for all of the return sorting windows (with a t-statistics of -7.31 for the past 1-week return). The result is mostly consistent with the results of Shen et al. (2020), who uses equal-weighted portfolio returns and does not apply a minimum market capitalization filter. Unlike the previous case, the reversal effect for Small and Liquid coins is driven by both legs of the Long-Short portfolio, with the Q1 portfolio significantly outperforming the others, and the Q5 portfolio earning negative returns. As it is unlikely

that the Q5 coins could be shorted, the practical usefulness of the strategy is dependent on our ability to buy the Q1 coins which may be complicated by their low liquidity.

Table 4 – High-Momentum/Reversal – Average quintile portfolio returns and t-statistics

High-momentum/reversal - Large and Liquid cryptocurrencies							
variable	metric	Q1	Q2	Q3	Q4	Q5	Q5-Q1
hmom_1W	mean	-0.70%	0.96%	1.28%	1.33%	1.98%	2.68%
	t-stat	-0.759	0.761	1.101	1.150	1.619	4.933
hmom_2W	mean	-0.43%	0.48%	1.12%	1.52%	2.14%	2.57%
	t-stat	-0.456	0.440	0.934	1.253	1.693	4.383
hmom_4W	mean	-0.38%	0.44%	0.95%	1.41%	2.22%	2.60%
	t-stat	-0.421	0.411	0.792	1.163	1.778	4.745
hmom_12W	mean	0.70%	0.41%	0.80%	1.24%	1.70%	1.00%
	t-stat	0.658	0.423	0.756	0.988	1.422	1.935
hmom_26W	mean	0.92%	0.67%	0.84%	0.70%	1.49%	0.57%
	t-stat	0.891	0.686	0.746	0.643	1.333	1.185
High-momentum/reversal - Small or Illiquid cryptocurrencies							
variable	metric	Q1	Q2	Q3	Q4	Q5	Q5-Q1
hmom_1W	mean	4.17%	1.00%	0.78%	0.64%	0.50%	-3.67%
	t-stat	3.554	0.952	0.685	0.570	0.426	-9.032
hmom_2W	mean	3.71%	1.27%	0.71%	0.78%	0.62%	-3.09%
	t-stat	3.429	1.112	0.668	0.666	0.507	-6.952
hmom_4W	mean	3.77%	0.98%	0.82%	0.68%	0.79%	-2.98%
	t-stat	3.353	0.898	0.786	0.567	0.666	-6.640
hmom_12W	mean	3.53%	1.45%	0.91%	0.74%	0.32%	-3.21%
	t-stat	3.102	1.521	0.842	0.636	0.278	-6.911
hmom_26W	mean	3.52%	1.48%	1.05%	0.68%	0.30%	-3.21%
	t-stat	2.868	1.508	1.016	0.670	0.277	-5.887

Note: The table shows average weekly returns (mean) and Newey-West t-statistics (t-stat) of 5 quintile portfolios (Q1 to Q5) and the Q5-Q1 long-short portfolio, with the sorts performed based on the distance of the previous week close price from the highest price computed over the past 1-week, 2-weeks, 4-weeks, 12-weeks or 26-weeks respectively, over the period of 2017/06-2022/12. Instances with t-statistic>2 are shown in bold. The upper table shows the results for Large and Liquid cryptocurrencies with previous-week market capitalization >=50 mil. USD and previous-week trading volume >=5 mil. USD. The lower table shows the results for small cryptocurrencies with market capitalization < 50 mil. USD (but still >1 mil. USD) or volume < 5 mil. USD.

We can see from **Table 4** that the Large and Liquid cryptocurrencies exhibit highly statistically significant high-momentum for sorting windows of up to 4 weeks (with a t-statistic of 4.93 for the 1-week return sorts). The high-momentum effect is not only significantly stronger than the one observed for the standard momentum, but seems to be also far more robust, as the quintile portfolio returns are monotonically increasing over quintiles (for 1W to 4W past returns). Both of the legs of the long-short portfolio thus contribute to the excess portfolio returns, and the practical utilization of the strategy is thus not dependent only on our ability to short the Q1 coins, but its profits may be partially utilized also by the purchase of the Q5 coins.

For the Small and Illiquid cryptocurrencies, we observe highly statistically significant reversal effect, which for the 1-week sorting period exceeds the significance of the standard momentum (t-statistic of -9.03). While the profits of the Long-Short strategy seem to again be driven by both of the legs, the majority of them comes from the long leg and purchase of the Q1 coins (t-statistic 3.55). As in the case of the standard momentum, the purchase of these coins for the reported closing price may be complicated by the low liquidity of the analysed coins as we did not use any filter on the trading

volumes (for the sake of consistency with other studies). The issues of practical utilization of the high-reversal strategy will be discussed in more detail in section 3.3 where we apply alternative dataset filtering rules.

3.2. Time-series regressions

In the next step, we compute the time-series regressions of our quantile portfolio returns on a series of previously published market factors.

As a first benchmark model we use the Bitcoin CAPM:

$$R_{q,t} = \alpha_{BTC} + \beta_{BTC} R_{BTC,t} + \varepsilon_{q,t} \quad (3)$$

Where $R_{q,t}$ denotes the return of the respective quintile portfolio in week t , $R_{BTC,t}$ is the return of Bitcoin in week t , $\varepsilon_{q,t}$ is the model residual and α_{BTC} and β_{BTC} are the model parameters.

We decided to use a Bitcoin CAPM instead of cryptocurrency market CAPM (Liu et al., 2022, Shen et al., 2020) as we view it more appropriate for our size-sorted dataset, and also because it will allow us to directly assess to what degree could the quantile portfolios be used to earn profits by replacing the short-leg of the long-short portfolio with a position in Bitcoin.

As a second benchmark model (for the high-momentum strategy only), we use the 3-Factor model inspired by Liu et al. (2022) and Shen et al. (2020) with a Bitcoin factor, size factor and 1-week momentum factor:

$$R_{q,t} = \alpha_{BTC} + \beta_{BTC} R_{BTC,t} + \beta_{MC} R_{MC,t} + \beta_{M1W} R_{M1W,t} + \varepsilon_{q,t} \quad (4)$$

where $R_{q,t}$ denotes the return of the respective quintile portfolio in week t , $R_{BTC,t}$ is the return of Bitcoin in week t , $R_{MC,t}$ is the return of Q1-Q5 portfolio sorted by previous-week market capitalization, and $R_{M1W,t}$ is the return of Q1-Q5 portfolio sorted by previous-week return.

The time-series regression tests are performed for the 1-week momentum strategies only as the results for the other sorting horizons are very similar.

Table 5 – 1W Momentum/Reversal – Realized alphas with respect to BTC CAPM

Momentum/Reversal - Large and Liquid cryptocurrencies						
factor	quantile	meanx	meanR	tStat	alphaBTC	tStatBTC
mom_1W	1	-0.169	-0.20%	-0.203	-1.48%	-1.980
mom_1W	2	-0.065	1.09%	1.035	-0.20%	-0.261
mom_1W	3	-0.016	1.44%	1.290	0.12%	0.163
mom_1W	4	0.044	1.33%	1.138	0.06%	0.068
mom_1W	5	0.245	1.15%	0.856	-0.17%	-0.180
mom_1W	5-1		1.36%	2.332	1.31%	2.603
Momentum/Reversal - Small or Illiquid cryptocurrencies						
factor	quantile	meanx	meanR	tStat	alphaBTC	tStatBTC
mom_1W	1	-0.279	3.44%	3.206	2.27%	2.611
mom_1W	2	-0.096	1.17%	1.079	0.02%	0.018
mom_1W	3	-0.026	1.47%	1.263	0.30%	0.304
mom_1W	4	0.051	1.45%	1.208	0.26%	0.245
mom_1W	5	0.330	-0.47%	-0.407	-1.63%	-1.694
mom_1W	5-1		-3.91%	-7.309	-3.90%	-7.590

Note: The table shows average weekly returns (meanR), realized BTC CAPM alphas (alphaBTC) and their respective Newey-West t-statistics (tStat and tStatBTC) of 5 quintile portfolios (Q1 to Q5) and the Q5-Q1 long-short portfolio, with the sorts performed based on the past 1-week returns in the period of 2017/06-2022/12. Instances with t-statistic>2 are shown in bold. The upper table shows the results for Large and Liquid cryptocurrencies with previous-week market capitalization >=50 mil. USD and previous-week trading volume >=5 mil. USD. The lower table shows the results for small or illiquid cryptocurrencies with market capitalization < 50 mil. USD (but still >1 mil. USD) or volume < 5 mil.

We can see from **Table 5** that the returns of the long-short strategy based on 1-week momentum remain statistically significant even when adjusted for their sensitivity to Bitcoin. For the Large and Liquid coins, the returns of the quantile portfolios (Q1 to Q5) remain statistically insignificant, indicating that investing in any of the quantile portfolios and hedging the investment with a position in Bitcoin does not represent a reliable strategy. In the case of the Small and Illiquid coins, the Q1 portfolio provides a statistically significant positive alpha (t-statistic 2.61) which indicates that investing in the Q1 coins and hedging the position with Bitcoin could be a profitable strategy under the assumption that the Q1 coins could be purchased for their end-of-week closing prices.

Table 6 – 1W High-Momentum/Reversal – Alphas with respect to BTC CAPM and the 3-F model

High-Momentum/Reversal - Large and Liquid cryptocurrencies								
factor	quantile	meanx	meanR	tStat	alphaBTC	tStatBTC	alpha3F	tStat3F
hmom_1W	1	-0.308	-0.70%	-0.759	-1.95%	-2.722	-1.79%	-3.319
hmom_1W	2	-0.173	0.96%	0.761	-0.40%	-0.441	-0.29%	-0.441
hmom_1W	3	-0.132	1.28%	1.101	-0.04%	-0.044	0.01%	0.021
hmom_1W	4	-0.099	1.33%	1.150	0.06%	0.074	-0.06%	-0.091
hmom_1W	5	-0.052	1.98%	1.619	0.69%	0.826	0.43%	0.655
hmom_1W	5-1		2.68%	4.933	2.64%	5.610	2.22%	5.720
High-Momentum/Reversal - Small or Illiquid cryptocurrencies								
factor	quantile	meanx	meanR	tStat	alphaBTC	tStatBTC	alpha3F	tStat3F
hmom_1W	1	-0.559	4.17%	3.554	3.01%	3.052	1.33%	1.534
hmom_1W	2	-0.251	1.00%	0.952	-0.22%	-0.269	-1.35%	-1.604
hmom_1W	3	-0.173	0.78%	0.685	-0.41%	-0.443	-1.61%	-1.790
hmom_1W	4	-0.117	0.64%	0.570	-0.55%	-0.579	-1.03%	-1.401
hmom_1W	5	-0.054	0.50%	0.426	-0.57%	-0.546	-1.09%	-1.651
hmom_1W	5-1		-3.67%	-9.032	-3.58%	-10.137	-2.43%	-4.373

Note: The table shows average weekly returns (meanR), realized BTC CAPM alphas (alphaBTC), 3-factor model alphas (alpha3F) and their respective Newey-West t-statistics (tStat, tStatBTC and tStat3F) of 5 quintile portfolios (Q1 to Q5) and the Q5-Q1 long-short portfolio, with the sorts performed based on the distance of the previous week close price from the highest price computed over the past 1-week in the period of 2017/06-2022/12. Instances with t-statistic > 2 are shown in bold. The upper table shows the results for Large and Liquid cryptocurrencies with previous-week market capitalization >= 50 mil. USD and previous-week trading volume >= 5 mil. USD. The lower table shows the results for small or illiquid cryptocurrencies with market capitalization < 50 mil. USD (but still > 1 mil. USD) or volume < 5 mil.

We can see from **Table 6** that the returns of the long-short high-momentum strategy remain statistically significant even when adjusted for their sensitivity the returns of Bitcoin, as well as to the returns of the factor-mimicking portfolios in the 3-factor model. The returns of the 1-week high-momentum strategy are thus not fully explained by the returns of the standard 1-week momentum. This holds especially for the Large and Liquid coins, whose alpha with respect to the 3-factor model is even more significant (t-stat = 5.72) than the mean return of the long-short strategy (t-stat = 4.93). For the Small and Illiquid coins, the magnitude of the alpha drops, when compared with the mean return (from -3.67% to -2.43%) but remains highly statistically significant (t-stat = -4.373).

Regarding the quantile portfolios, for the Large and Liquid coins, only the Q1 negative alphas happen to be significant. The strategy of purchasing the Q5 coins and hedging the position with bitcoin, while earning a positive profit of 0.69% per week historically, cannot be viewed as reliable (t-stat = 0.83). Utilization of the high-momentum anomaly is thus dependent on shorting the Q1 coins.

For the Small and Illiquid coins, we can see a statistically significant Bitcoin CAPM alpha of 3.01% per week (t-stat = 3.052), indicating that the purchase of Q1 coins and hedging of the position with Bitcoin could be a potentially viable strategy, under the assumption that these coins could be purchased for their end-of-week closing prices.

3.3. Alternative dataset definitions

The setting of the market capitalization threshold to 50 mil. USD and the weekly USD volume threshold to 5 mil. USD in the previous sections was mostly expert based. The goal of this section is to show the robustness of the results with respect to alternative thresholds. Additionally, we will strive to answer the question of whether the market capitalization or rather the USD trading volumes drive the observed momentum and reversal effects.

Table 7 – Momentum – Long-Short portfolio returns for alternative size and volume filters

Panel A: Momentum / Reversal - High market capitalization & High USD volume					
factor	metric	M>1M, V>100K	M>10M, V>1M	M>50M, V>5M	M>250M, V>25M
mom_1W	mean	-0.12%	1.44%	1.36%	1.82%
	t-stat	-0.216	2.075	2.332	2.816
mom_2W	mean	-0.15%	1.77%	1.44%	1.70%
	t-stat	-0.304	2.479	2.686	2.428
mom_4W	mean	-0.50%	1.13%	0.84%	1.34%
	t-stat	-1.054	1.950	1.570	2.084
mom_12W	mean	-1.38%	-0.69%	-0.88%	-0.28%
	t-stat	-3.501	-1.376	-1.862	-0.421
mom_26W	mean	-1.22%	-0.46%	-0.71%	-0.59%
	t-stat	-3.564	-1.340	-1.914	-0.946
Panel B: Momentum / Reversal - High market capitalization & Low USD volume					
factor	metric	M>1M, V<100K	M>10M, V<1M	M>50M, V<5M	M>250M, V<25M
mom_1W	mean	-3.06%	0.54%	1.19%	2.08%
	t-stat	-5.928	0.799	1.719	2.794
mom_2W	mean	-2.58%	0.74%	1.07%	1.53%
	t-stat	-5.118	1.194	1.937	2.367
mom_4W	mean	-2.58%	0.48%	0.44%	1.48%
	t-stat	-5.701	0.875	0.772	2.232
mom_12W	mean	-3.07%	-1.21%	-1.26%	-0.54%
	t-stat	-7.471	-2.310	-2.210	-0.967
mom_26W	mean	-2.70%	-0.99%	-1.08%	-1.07%
	t-stat	-6.245	-2.230	-2.161	-1.735
Panel C: Momentum / Reversal - Low market capitalization & High USD volume					
factor	metric	M<1M, V>100K	M<10M, V>1M	M<50M, V>5M	M<250M, V>25M
mom_1W	mean	-1.15%	0.23%	1.07%	1.25%
	t-stat	-2.256	0.385	1.493	1.558
mom_2W	mean	-1.07%	0.24%	1.03%	1.09%
	t-stat	-2.181	0.412	1.618	1.743
mom_4W	mean	-1.46%	0.01%	0.50%	0.09%
	t-stat	-3.487	0.017	1.018	0.155
mom_12W	mean	-2.25%	-1.31%	-0.90%	-1.19%
	t-stat	-5.358	-3.111	-1.638	-2.130
mom_26W	mean	-2.35%	-1.26%	-0.39%	-1.11%
	t-stat	-5.473	-3.538	-1.084	-2.035

Note: The table shows the performance of the Q5-Q1 long-short strategy based on momentum portfolio sorts for alternative dataset filters applied on the previous-week market capitalization (M) or the previous-week USD trading volumes (V). The dataset filters in columns range from 1 million to 250 million for the market capitalization and from 100 thousand to 25 million for the USD trading volume. The rows show the mean Q5-Q1 portfolio returns (mean) and Newey-West t-statistics (t-stat) of strategies based alternatively on the 1-week, 2-week, 4-week, 12-week and 26-week momentum).

We can see from Panel A of **Table 7** that the coins with market capitalization above 10 mil. USD and last-week trading volume above 1 mil. USD already exhibit statistically significant short-term momentum effects (t-stat 2.08 for 1-week returns and 2.48 for 2-week returns), which further increases once we move to the larger and more liquid coins (up to t-stat = 2.86 for the coins with over 250 mil. USD market capitalization and over 25 mil. USD trading volume).

Panel B further extends the results for coins with high market capitalization but relatively low trading volumes. Comparing the first column in Panel A and Panel B we can see that the highly significant short-term reversal effects happen to be driven mostly by the coins with very low weekly trading volumes, as while the coins with above 1 mil. USD market cap and above 100 thousand USD trading volumes exhibit no statistically significant short-term reversal (1-week t-stat = -0.21), the coins with above 1 mil. USD market cap and below 100 thousand USD trading volumes exhibit a highly statistically significant short-term reversal (1-week t-stat = -5.93). To observe statistically significant momentum effects in Panel B, we then need to move up to the group of largest coins with market capitalization above 250 mil. USD, while for the more actively traded coins in Panel A we could observe significant momentum effects even for coins with above 10 mil. market capitalization.

Panel C takes the opposite view of looking at the coins with relatively low market capitalization but high trading volumes. Comparing column 1 in Panel C with the one in Panel B, we can see that the statistical significance of the short-term reversal significantly decreased (1-week t-stat = -2.26 in Panel C, compared to a 1-week t-stat = -5.93 in Panel B), showing that once we filter out the most illiquid coins (with last week USD trading volume under 100 thousand USD), the short-term reversal rapidly decreases. The short-term reversal effect may thus be driven mostly by low liquidity issues which could make it difficult to realize its reported returns in practice. Surprisingly, when filtering based on volume only, in the remaining columns in Panel C, we do not observe any statistically significant momentum effects at all. While the short-term reversal of small illiquid coins was driven mostly by the low volume effect, the short-term momentum of large and liquid coins seems to be driven mostly by the large market capitalization effect.

In **Table 8** we show the long-term portfolio returns of the high-momentum based on different dataset filters. We can see from Panel A that any even the lowest filters on volume and market capitalization (above 1 mil. market capitalization and above 100 thousand trading volumes) result in a short-term momentum effect, which becomes highly statistically significant (t-stat > 4) for the higher filters (above 10 mil. market capitalization and above 1 mil. trading volumes).

Panel B shows that by selecting the coins above 1 mil. market capitalization, but with under 100 thousand USD trading volumes, results in a significant high-reversal effect (t-stat of -6.98 for the 1-week returns), which changes into a momentum effect once the market capitalization filter is increased above 1 million USD.

Panel C shows that by selecting the coins below 1 mil. market capitalization, but with higher than 100 thousand USD trading volumes, the strength of the short-term reversal significantly decreases (t-stat of -2.20 for the 1-week returns) compared to the results in Panel B. Coins with above 1 million market capitalization exhibit the high-momentum effect even with no filter on market capitalization, although a slightly weaker one than in Panel B and Panel A where the filter on market capitalization was applied.

The results confirm the hypothesis that the short-term reversal is driven mostly by the low trading volumes, while the short-term momentum is driven mostly by high market capitalizations, with the volumes playing role especially for the medium-sized coins where they can contribute to the momentum effect.

Table 8 – High-Momentum – Long-Short portfolio returns for alternative size and volume filters

Panel A: High Momentum / Reversal - High market capitalization & High USD volume					
factor	metric	M>1M, V>100K	M>10M, V>1M	M>50M, V>5M	M>250M, V>25M
hmom_1W	mean	0.40%	2.07%	2.68%	2.26%
	t-stat	0.883	4.704	4.933	3.584
hmom_2W	mean	0.47%	2.06%	2.57%	2.56%
	t-stat	1.141	4.224	4.383	4.367
hmom_4W	mean	0.42%	2.11%	2.60%	2.35%
	t-stat	1.050	3.858	4.745	3.905
hmom_12W	mean	-0.33%	1.04%	1.00%	1.29%
	t-stat	-0.823	1.977	1.935	2.040
hmom_26W	mean	-0.73%	0.61%	0.57%	0.71%
	t-stat	-1.716	1.160	1.185	1.119
Panel B: High Momentum / Reversal - High market capitalization & Low USD volume					
factor	metric	M>1M, V<100K	M>10M, V<1M	M>50M, V<5M	M>250M, V<25M
hmom_1W	mean	-2.73%	1.00%	2.48%	2.54%
	t-stat	-6.981	2.399	4.640	4.133
hmom_2W	mean	-2.24%	1.23%	2.34%	2.98%
	t-stat	-5.723	2.695	3.991	4.979
hmom_4W	mean	-2.07%	1.39%	2.25%	3.23%
	t-stat	-4.773	2.520	4.034	4.830
hmom_12W	mean	-2.58%	0.30%	0.63%	1.87%
	t-stat	-6.436	0.625	1.161	3.025
hmom_26W	mean	-2.79%	-0.10%	0.35%	1.12%
	t-stat	-5.807	-0.213	0.718	1.862
Panel C: High Momentum / Reversal - Low market capitalization & High USD volume					
factor	metric	M<1M, V>100K	M<10M, V>1M	M<50M, V>5M	M<250M, V>25M
hmom_1W	mean	-0.96%	1.08%	2.00%	2.44%
	t-stat	-2.197	2.492	4.036	3.458
hmom_2W	mean	-0.92%	1.06%	1.99%	2.13%
	t-stat	-2.402	2.435	3.905	3.293
hmom_4W	mean	-0.91%	1.03%	1.84%	2.14%
	t-stat	-2.431	2.284	3.074	3.516
hmom_12W	mean	-1.52%	0.06%	0.72%	0.81%
	t-stat	-3.740	0.133	1.307	1.231
hmom_26W	mean	-2.03%	-0.43%	0.35%	0.84%
	t-stat	-4.290	-0.953	0.640	1.351

Note: The table shows the performance of the Q5-Q1 long-short strategy based on high-momentum portfolio sorts for alternative dataset filters applied on the previous-week market capitalization (M) or the previous-week USD trading volumes (V). The dataset filters in columns range from 1 million to 250 million for the market capitalization and from 100 thousand to 25 million for the USD trading volume. The rows show the mean Q5-Q1 portfolio returns (mean) and Newey-West t-statistics (t-stat) of strategies based alternatively on the 1-week, 2-week, 4-week, 12-week and 26-week momentum).

4. Conclusion

The goal of our study is to analyse how cryptocurrency size and trading volumes impact the momentum and reversal dynamics of their returns. We show that the previously reported short-term reversal (Shen et al., 2020, Kozłowski et al., 2020), while highly statistically significant (NW t-stat of -7.31) for the 1-week returns, is mostly a feature of small and illiquid coins with low trading volumes. Large, actively traded coins exhibit statistically significant short-term momentum (with NW t-statistics of 2.33 and 2.68 for the 1-week and 2-week returns respectively), consistent with the result of Liu et al. (2022) and Dobrynskaya (2023). Returns beyond 4-weeks exhibit a reversal effect even for the large and liquid coins, although insignificant one (NW t-stat of -1.91 for 26-week returns), supporting the explanation of the momentum by the initial-underreaction / delayed-overreaction hypothesis.

As an alternative measure of momentum, we test the high-momentum, measured as the distance of the end of week closing price from the previous k-week high (George and Hwang, 2004), which was not applied to the cryptocurrency market before. We show that it represents a superior predictor of future cryptocurrency returns, with even stronger discrepancy between the large and liquid coins and the small and illiquid ones. The 1-week high-momentum predicts positively future returns of large and liquid coins (NW t-stat = 4.93) and negatively the returns of small and illiquid coins (NW t-stat = -9.03).

We further show that while the short-term reversal, observed for small and illiquid coins, is driven mostly by the long leg of the relationship (buying the coins that underperformed), the short-term momentum, observed for large and liquid coins, is driven mostly by the short leg of the relationship (shorting the coins that outperformed). This could make it challenging to utilize the reversal and momentum effects in practice as they depend either on the purchase of coins that are illiquid (to utilize the reversal) or on the shorting of coins that are liquid (to utilize the momentum), which might be, in spite of their liquidity, potentially problematic on the cryptocurrency market.

In the analysis of alternative size and volume filters we show that the results are highly robust to the definitions of large and liquid coins vs. small and illiquid coins. Furthermore, we show that the short-term reversal happens to be driven mostly by coins with very low trading volumes (under 100 thousand USD in the last week), which might be difficult to purchase for the reported closing price. The short-term momentum, on the other hand, is mostly driven by coins with large market capitalization, as robust even for the largest and most liquid group of coins (above 250 mil. USD market capitalization and above 25 mil. USD trading volumes).

In our future research we will focus on analysing the impact of cryptocurrency size and trading volumes on another anomalies reported in the cryptocurrency pricing literature as they may too be influenced by the cryptocurrency size and liquidity. Additionally, while the observed short-term reversal seems to be driven mostly by the low trading volumes and low liquidity, the true source of the momentum effect remains unknown, and it is possible that market capitalization (supported by volume for medium-sized coins) is still not the ideal proxy for it. Analysis of other potential predictors of the momentum affect, such as attention, sentiment or institutional trading activity may thus prove worthwhile as well.

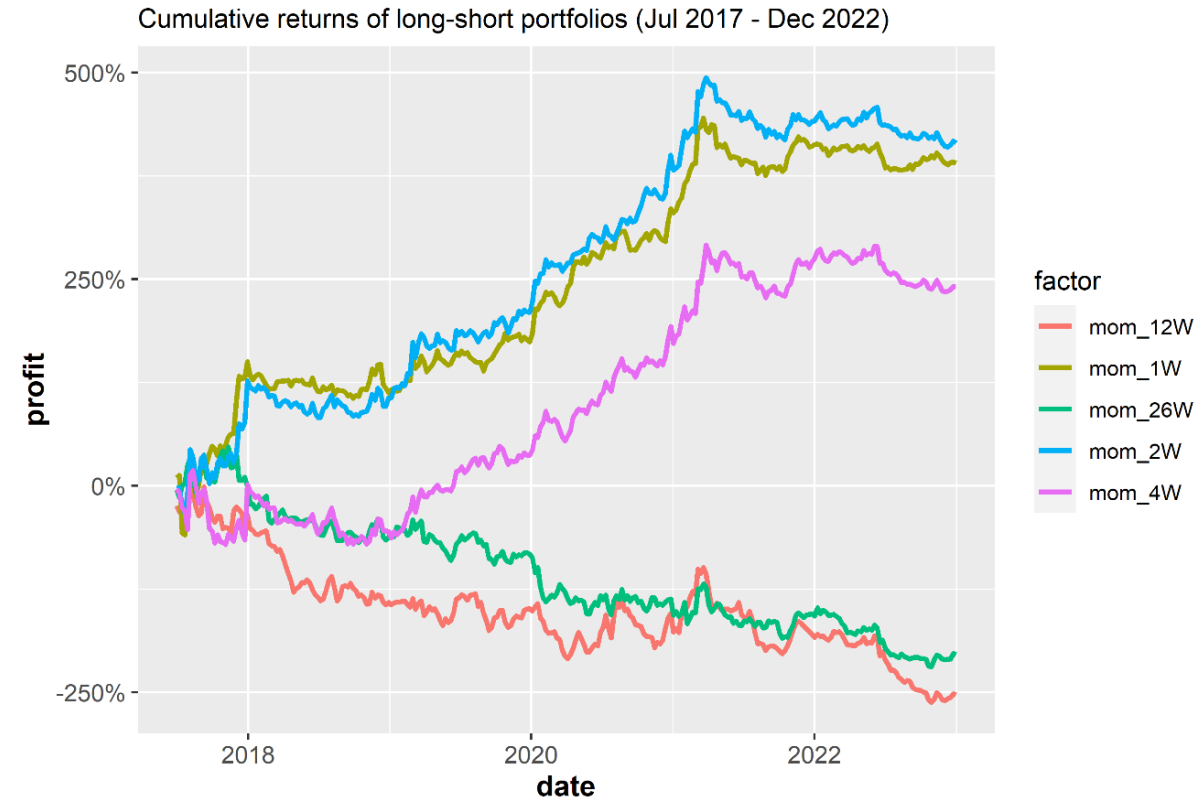
5. References

- [1] Dobrynskaya, V. (2023). "Cryptocurrency Momentum and Reversal". *Journal of Alternative Investments*, forthcoming
- [2] George, T. J., Hwang, C.-Y. (2004). "The 52-Week High and Momentum Investing". *The Journal of Finance*. 59 (5): 2145-2176
- [3] Grobys, K., Sapkota, N. (2019). "Cryptocurrencies and momentum". *Economic Letters*. 180: 6-10
- [4] Jia, B., Goodell, J. W., Shen, D. (2022). "Momentum or reversal: Which is the appropriate third factor for cryptocurrencies?", *Finance Research Letters*, 45, 102139
- [5] Jia, Y., Liu, Y., Yan, S. (2021). "Higher moments, extreme returns, and cross-section of cryptocurrency returns". *Finance Research Letters*, 39, 101536
- [6] Kozlowski, S. E., Puelo, M. R., Zhou, J. (2020). "Cryptocurrency return reversals". *Applied Economics Letters*. 28 (11): 887-893
- [7] Lervik, T. (2022). "Cryptocurrency returns and the volatility of liquidity". *Finance Research Letters*. 44, 102031
- [8] Li, Y., Urquhart, A., Wang, P., Zhang, W. (2021). "MAX momentum in cryptocurrency markets". *International Review of Financial Analysis*. 77, 101829
- [9] Liebi, L. J. (2022). "Is there a value premium in cryptoasset markets?". *Economic Modelling*. 109, 105777
- [10] Liu, W., Liang, X., Cui, G. (2020). "Common risk factors in the returns on cryptocurrencies". *Economic Modelling*. 86: 299-305
- [11] Liu, Y., Tsyvinski, A., Wu, X. (2022). "Common Risk Factors in Cryptocurrency". *The Journal of Finance*. 77 (2): 1133-1177
- [12] Long, H., Zaremba, A., Demir, E., Szczygielski, J. J., Vasenin, M. (2020). "Seasonality in the Cross-Section of Cryptocurrency Returns". *Finance Research Letters*. 35, 101566
- [13] Shen, D., Urquhart, A., Wang, P. (2020). "A three-factor pricing model for cryptocurrencies". *Finance Research Letters*. 34, 101248
- [14] Zaremba, A., Bilgin, M. H., Long, H., Mercik, A., Szczygielski, J. J. (2021). "Up or down? Short-term reversal, momentum, and liquidity effects in cryptocurrency markets". *International Review of Financial Analysis*. 78, 101908
- [15] Zhang, W., Li, Y. (2020). "Is idiosyncratic volatility priced in cryptocurrency markets?". *Research in International Business and Finance*. 54, 101252
- [16] Zhang, W., Li, Y. (2023). "Liquidity risk and expected cryptocurrency returns". *International Journal of Finance and Economics*. 28(1): 472-492
- [17] Zhang, W., Li, Y., Xiong, X., Wang, P. (2021). "Downside risk and the cross-section of cryptocurrency returns". *Journal of Banking & Finance*. 133, 106246
- [18] Zhang, Z., Zhao, R. (2021). "Good Volatility, Bad Volatility, and the Cross Section of Cryptocurrency Returns". SSRN. No. 3910202

6. Appendices

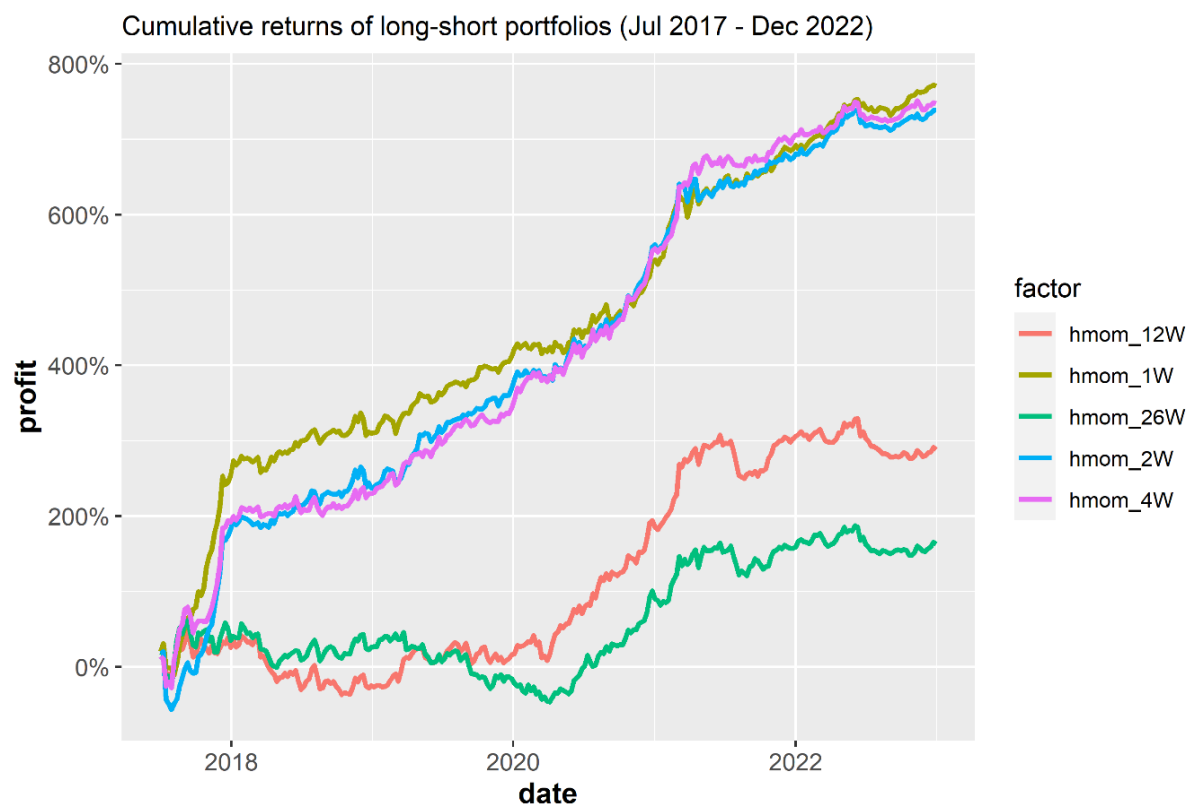
6.1. Appendix 1 – Cumulative long-short portfolio returns

Figure 1 – Momentum – Long-short portfolio returns – Large and Liquid coins



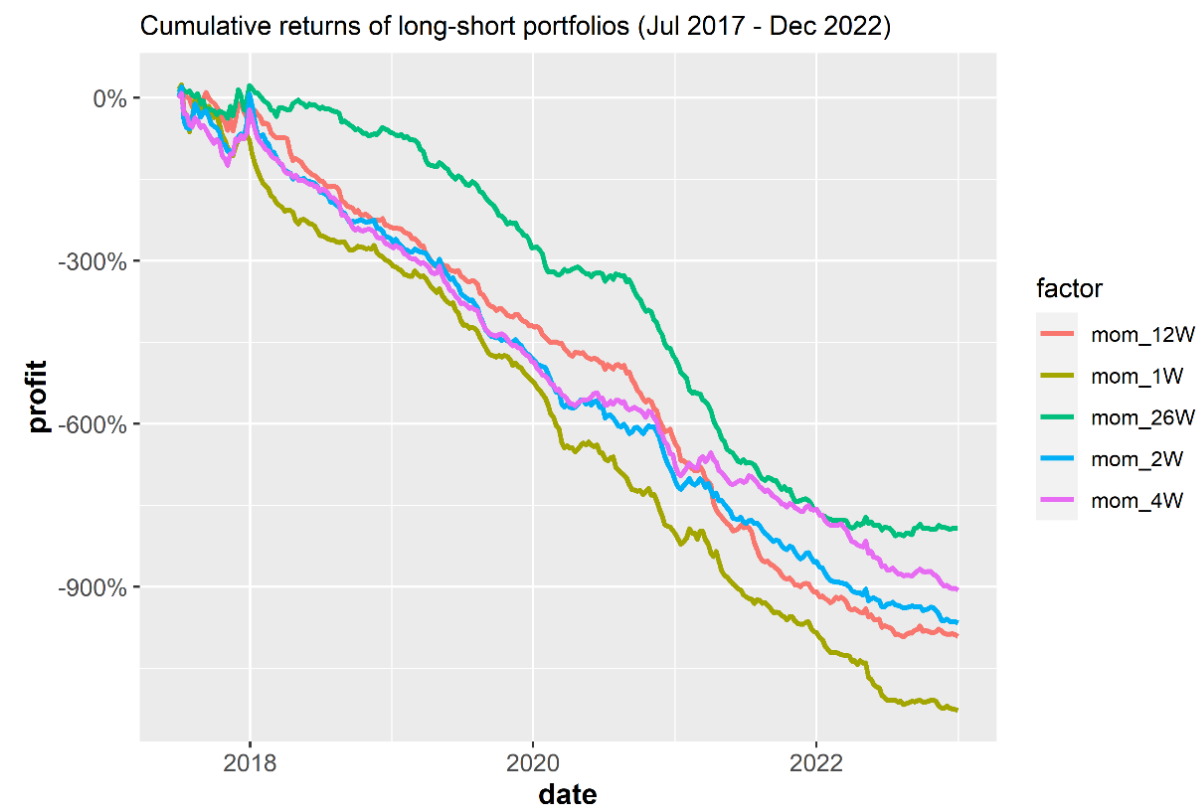
Note: The chart shows cumulative long-short portfolio returns (Q5-Q1) of cryptocurrency quantile portfolios sorted by their previous momentum (1-week up to 26-week) at the end of each week, in the period of 2017/06-2022/12. Only coins with previous-week market capitalization above 50 mil. USD and previous-week volume above 5 mil. USD are traded.

Figure 2 – High-Momentum – Long-short portfolio returns – Large and Liquid coins



Note: The chart shows cumulative long-short portfolio returns (Q5-Q1) of cryptocurrency quantile portfolios sorted by their previous high-momentum (1-week up to 26-week) at the end of each week, in the period of 2017/06-2022/12. Only coins with previous-week market capitalization above 50 mil. USD and previous-week volume above 5 mil. USD are traded.

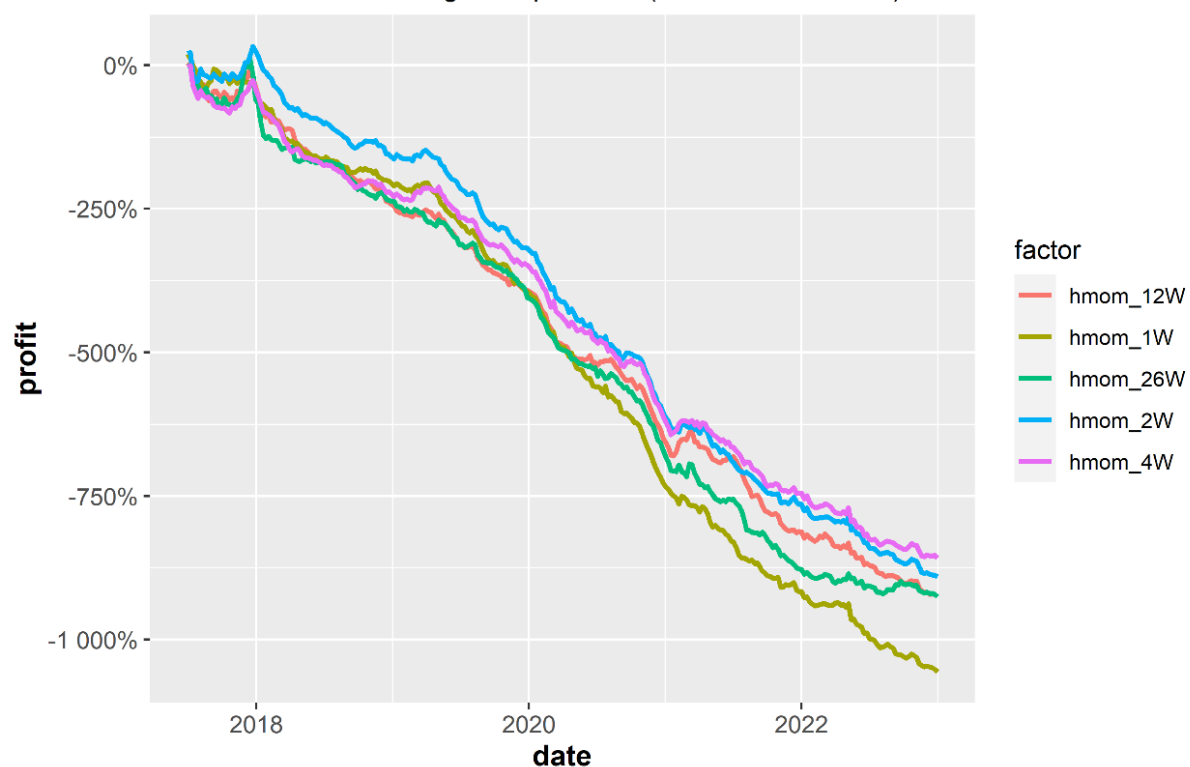
Figure 3 – Momentum – Long-short portfolio returns – Small and Illiquid coins



Note: The chart shows cumulative long-short portfolio returns (Q5-Q1) of cryptocurrency quantile portfolios sorted by their previous momentum (1-week up to 26-week) at the end of each week, in the period of 2017/06-2022/12. Only coins with previous-week market capitalization between 100 thousand USD and 50 mil. USD or previous-week market capitalization above 50 thousand USD but trading volume below 5 mil. USD are traded.

Figure 4 – High-Momentum – Long-short portfolio returns – Small and Illiquid coins

Cumulative returns of long-short portfolios (Jul 2017 - Dec 2022)



Note: The chart shows cumulative long-short portfolio returns (Q5-Q1) of cryptocurrency quantile portfolios sorted by their previous high-momentum (1-week up to 26-week) at the end of each week, in the period of 2017/06-2022/12. Only coins with previous-week market capitalization between 100 thousand USD and 50 mil. USD or previous-week market capitalization above 50 thousand USD but trading volume below 5 mil. USD are traded.

6.2. Appendix 2 – Manually excluded coins

Below is a list of stable coins, mirrored coins and wrapped coins that were manually removed from the dataset to increase its representativeness. This is done as the dynamics of stable coins and mirrored coins is likely to be different from the one observed for free-floating coins. The wrapped coins, on the other hand, would mostly just duplicate observations in our dataset. All of the removed coins are denoted based on their symbol, name and id, as reported by Coinmarketcap.com.

Table 9 - Manually excluded coins

symbol	name	id	symbol	name	id	symbol	name	id
FLL	Feellike	6410	VNDC	VNDC	4805	mQQQ	Mirrored Invesco QQQ Trust	8025
vUSDC	Venus USDC	7958	EURS	STASIS EURO	2989	mAAPL	Mirrored Apple	8001
BUSD	Binance USD	4687	nUSD-EXCL	Neutral Dollar	3975	mMSFT	Mirrored Microsoft	8017
vDAI	Venus DAI	8214	BGBP	Binance GBP Stable Coin	4186	mVIX	Mirrored ProShares VIX	8028
vUSDT	Venus USDT	7957	BITCNY	bitCNY	624	mIAU	Mirrored iShares Gold Trust	8024
vBUSD	Venus BUSD	7959	TOKEN	SuperNET (TOKEN)	612	mSLV	Mirrored iShares Silver Trust	8026
DAI	Dai	4943	XCHF	CryptoFranc	4075	mUSO	Mirrored United States Oil Fund	8027
USDC	USD Coin	3408	XSGD	XSGD	8489	mNFLX	Mirrored Netflix	8005
EDGT	Edgecoin	9466	XAUT	Tether Gold	5176	mAMZN	Mirrored Amazon	8016
USDT	Tether	825	CGO	Comtech Gold	20245	mTSLA	Mirrored Tesla	8004
TUSD	TrueUSD	2563	VEETH	Vether	5657	mBABA	Mirrored Alibaba	8006
USDP	Pax Dollar	3330	CCXX	CCX	5482	mTWTR	Mirrored Twitter	8018
HUSD	HUSD	4779	PAXG	PAX Gold	4705	WTRX	Wrapped TRON	18579
TRMB	TRMB	941	USDS	Stably USD	3719	WKAI	Wrapped KardiaChain	19080
USDD	USDD	19891	CGT	CACHE Gold	5719	WBTC	Wrapped Bitcoin	3717
CUSD	Celo Dollar	7236	PMGT	Perth Mint Gold Token	5203	WBNB	Wrapped BNB	7192
USDK	USDK	4064	EOSDT	EOSDT	4017	WEVER	Wrapped Everscale	20701
DLC	Diamond Launch	21849	VAI	Vai	7824	WVLX	Wrapped Velas	19076
OUSD	Origin Dollar	7189	USDS	Stronghold USD	3641	WNXM	Wrapped NXM	5939
ZUSD	ZUSD	8772	KRT	TerraKRW	5115	WCK	Wrapped Basic CryptoKitties	7473
FRAX	Frax	6952	USDx	USDx [Kava]	6651	WNCG	Wrapped NCG (Nine Chronicles Gold)	11222
USDJ	USDJ	5446	KT	Kuai Token	3691	WG0	Wrapped Gen-0 CryptoKitties	7472
FEI	Fei USD	8642	CONST	Constant	3739	WOA	Wrapped Origin Axie	7342
RSV	Reserve	6727	CTG	City Tycoon Games	22540	WVG0	Wrapped Virgin Gen-0 CryptoKitties	7471
LUSD	Liquity USD	9566	BITUSD	bitUSD	623			
GUSD	Gemini Dollar	3306	BITEUR	bitEUR	954			
MUSD	mStable USD	5747	WSD	White Standard	3003			
EUROC	Euro Coin	20641	EBASE	EURBASE	4815			
SUSD	sUSD	2927	GOLD	Digital Gold	4942			
ANCT	Anchor	4901	USTC	TerraClassicUSD	7129			
XTN	Neutrino Index	5068	1SG	1SG	3762			
RAI	Rai Reflex Index	8525	DGX	Digix Gold Token	2739			
CEUR	Celo Euro	9467	USDx	dForce USDx	4621			
SAI	Single Collateral DAI	2308	USDQ	USDQ	4020			
QC	Qcash	2319	BITGOLD	bitGold	778			
GYEN	GYEN	8771	BITSILVER	bitSilver	813			

FFA Working Paper Series

2019

1. Milan Fičura: Forecasting Foreign Exchange Rate Movements with k-Nearest-Neighbor, Ridge Regression and Feed-Forward Neural Networks.

2020

1. Jiří Witzany: Stressing of Migration Matrices for IFRS 9 and ICAAP Calculations.
2. Matěj Maivald, Petr Teplý: The impact of low interest rates on banks' non-performing loans.
3. Karel Janda, Binyi Zhang: The impact of renewable energy and technology innovation on Chinese carbon dioxide emissions.
4. Jiří Witzany, Anastasiia Kozina: Recovery process optimization using survival regression.

2021

1. Karel Janda, Oleg Kravtsov: Banking Supervision and Risk-Adjusted Performance in the Host Country Environment.
2. Jakub Drahokoupil: Variance Gamma process in the option pricing model.
3. Jiří Witzany, Ol'ga Pastiranová: IFRS 9 and its behaviour in the cycle: The evidence on EU Countries.

2022

1. Karel Janda, Ladislav Kristoufek, Binyi Zhang: Return and volatility spillovers between Chinese and U.S. Clean Energy Related Stocks: Evidence from VAR-MGARCH estimations.
2. Lukáš Fiala: Modelling of mortgage debt's determinants: The case of the Czech Republic.
3. Ol'ga Jakubíková: Profit smoothing of European banks under IFRS 9.
4. Milan Fičura, Jiří Witzany: Determinants of NMD Pass-Through Rates in Eurozone Countries.

5. Sophio Togonidze, Evžen Kočenda: Macroeconomic Responses of Emerging Market Economies to Oil Price Shocks: An Analysis by Region and Resource Profile.
6. Jakub Drahokoupil: Application of the XGBoost algorithm and Bayesian optimization for the Bitcoin price prediction during the COVID-19 period.
7. Karel Janda, Binyi Zhang: Econometric estimation of green bond premiums in the Chinese market.
8. Shahriyar Aliyev, Evžen Kočenda: ECB monetary policy and commodity prices.
9. Sophio Togonidze, Evžen Kočenda: Macroeconomic implications of oil-price shocks to emerging economies: a Markov regime-switching approach.

2023

1. Jiří Witzany, Milan Fičura: Machine Learning Applications for the Valuation of Options on Non-liquid Option Markets.
2. David Mazáček, Jiří Panoš: Key determinants of new residential real estate prices in Prague.
3. Milan Fičura: Impact of size and volume on cryptocurrency momentum and reversal.

All papers can be downloaded at: wp.ffu.vse.cz

Contact e-mail: ffawp@vse.cz



Faculty of Finance and Accounting, Prague University of Economics and Business, 2023

Winston Churchill Sq. 1938/4, CZ-13067 Prague 3, Czech Republic, ffu.vse.cz